Implementing Cloud Threat Detection and Automated Response with Microsoft Sentinel

Implementation Report | CloudScale Logistics | Microsoft Azure, East US

Analyst	Omajuwa Okotie
Client	CloudScale Logistics (mid-market freight and supply chain)
Engagement	Microsoft Sentinel SIEM and SOAR, two-week proof of concept
Primary region	East US
Frameworks	NIST Cybersecurity Framework, MITRE ATT&CK, SOC 2 Type II, ISO 27001

1. Executive Summary

CloudScale Logistics can now see and respond to threats across its entire Azure estate from one place, where before it had no unified view at all. Over a two-week proof of concept I deployed Microsoft Sentinel as a combined security information and event management (SIEM) and security orchestration, automation and response (SOAR) platform, connected the core Azure data sources into a single Log Analytics Workspace, enabled six analytics rules mapped to known attacker behaviour, and built an automated Logic Apps playbook that notifies the security team within seconds of a high-priority incident.

The headline outcome is a change in response capability. Before this engagement, a suspected credential compromise took over four hours to move from alert to containment decision, and all administrative activity was scattered across separate portals with no central record. After this engagement, one hundred percent of Azure control-plane actions are searchable from a single workspace, detections fire automatically every five minutes, and the security team receives a structured incident notification within seconds of an incident being raised, verified end to end on a real, naturally occurring incident rather than a simulated one. This moves the client from a reactive posture to a proactive one and directly addresses the ISO 27001 audit finding that triggered the project.

2. Project Context

CloudScale Logistics is a mid-market freight forwarding and supply-chain company serving over 200 enterprise clients, with annualised revenue in the $50 to $150 million range. It moved its core platforms, real-time freight tracking, inventory management, a customer portal, and payment processing, to Azure over the eighteen months before this engagement. The migration delivered the scalability the business wanted, but it also widened the attack surface across multiple subscriptions while the in-house team remained skilled mainly in on-premises infrastructure. Three problems followed from this.

- No unified visibility. Activity logs, Defender alerts, and VM events lived in separate places and were never consolidated, so an attacker could move between subscriptions without raising a timely alert.
- Manual, slow response. With no correlation and no automation, a small team of two analysts and a part-time CISO triaged hundreds of raw alerts by hand and escalated by email. One real incident, a suspected credential compromise in the payment-processing subscription, took more than four hours to reach a containment decision.
- An audit gap. ISO 27001 auditors issued a non-conformance finding because the client could not demonstrate a documented detection and response process for its cloud systems.

The goal of the proof of concept was therefore narrow and concrete: prove that a centralised, automated detection and response capability can be stood up quickly in Azure, and produce the evidence an auditor needs to close the finding.

3. Solution Architecture

The design follows three pillars that map to the Detect and Respond phases of the NIST Cybersecurity Framework: collect the data, identify the threats, and respond automatically. Data flows in one direction through the platform, which makes the system easy to reason about and easy to audit.

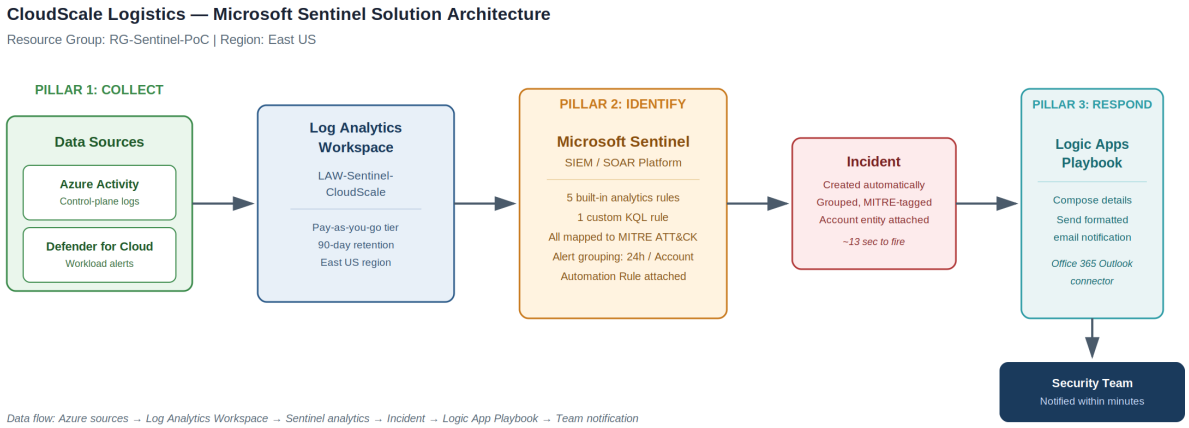


Figure 1. End-to-end architecture, from Azure data sources through to security team notification. All resources are deployed in resource group RG-Sentinel-PoC, East US.

In plain terms: Azure Activity logs and Microsoft Defender for Cloud feed raw security telemetry into the Log Analytics Workspace named LAW-Sentinel-CloudScale. Microsoft Sentinel sits on top of that workspace and runs analytics rules written in Kusto Query Language (KQL) against the incoming data. When a rule matches, Sentinel raises an incident. The incident triggers a Logic App playbook, which formats the details and emails them to the security team. Every stage leaves a record in the workspace, which is what gives the auditor a full trail.

3.1 Component Roles

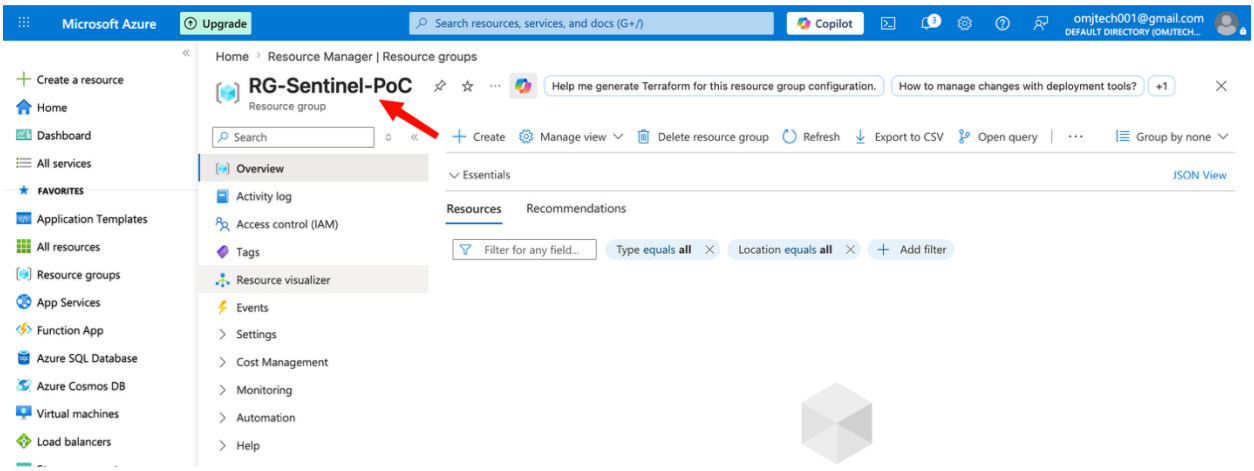
Component	Role in the Design
Log Analytics Workspace	Central store and query engine for all security telemetry. Named LAW-Sentinel-CloudScale, deployed in resource group RG-Sentinel-PoC.
Microsoft Sentinel	The SIEM and SOAR layer. Runs analytics rules, manages incidents, hosts playbooks.
Azure Activity logs	Control-plane data source. Records who did what, when, and from where across the subscription.
Microsoft Defender for Cloud	Workload data source. Supplies enriched alerts for VMs, storage and other resources, mapped to MITRE ATT&CK. The free Foundational CSPM tier was used; paid plans were deliberately deferred (see Section 7).
Azure Logic Apps	The automation engine behind the playbook. Low-code, fires on incident creation.
Kusto Query Language (KQL)	The query language used to write detection rules and verify data.

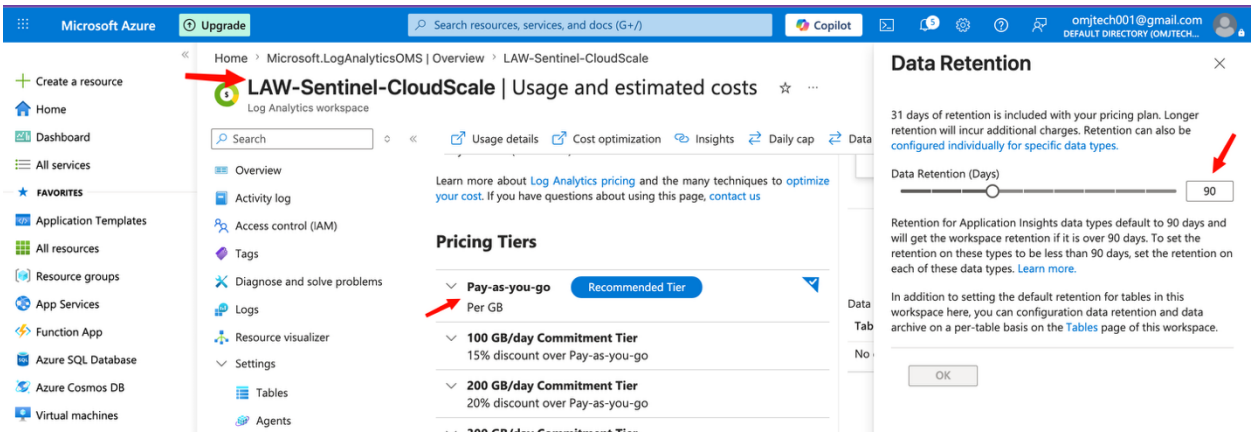
4. Implementation

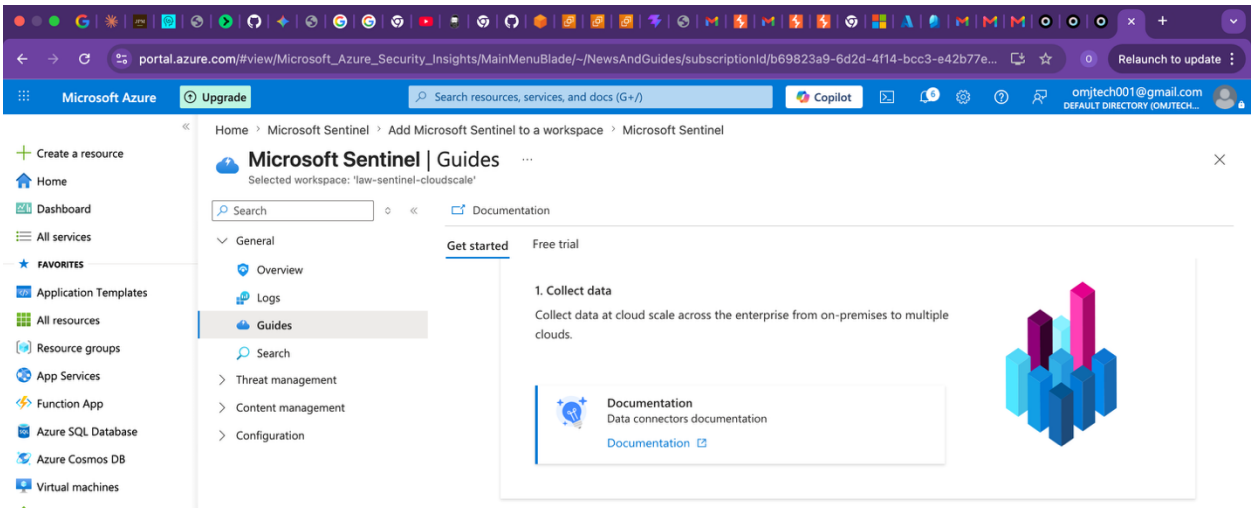
The build followed the three pillars in order, because each one depends on the last. You cannot detect without data, and you cannot automate a response to an incident that detection never raises. Resource names below are the real names used, so that this section can be used to rebuild the environment exactly.

4.1 Foundation (Week 1, Day 1)

I created a single resource group, RG-Sentinel-PoC, in East US to hold every resource. Keeping all project resources in one group makes cost tracking and clean-up simple, which matters for a time-boxed proof of concept. Inside it I deployed the Log Analytics Workspace LAW-Sentinel-CloudScale on the pay-as-you-go tier with retention set to ninety days. Ninety days is the floor required to satisfy the client's compliance obligation; I recommend one hundred and eighty days for production if budget allows. I then enabled Microsoft Sentinel on that workspace, which activates the SIEM and SOAR features on top of the raw data store.





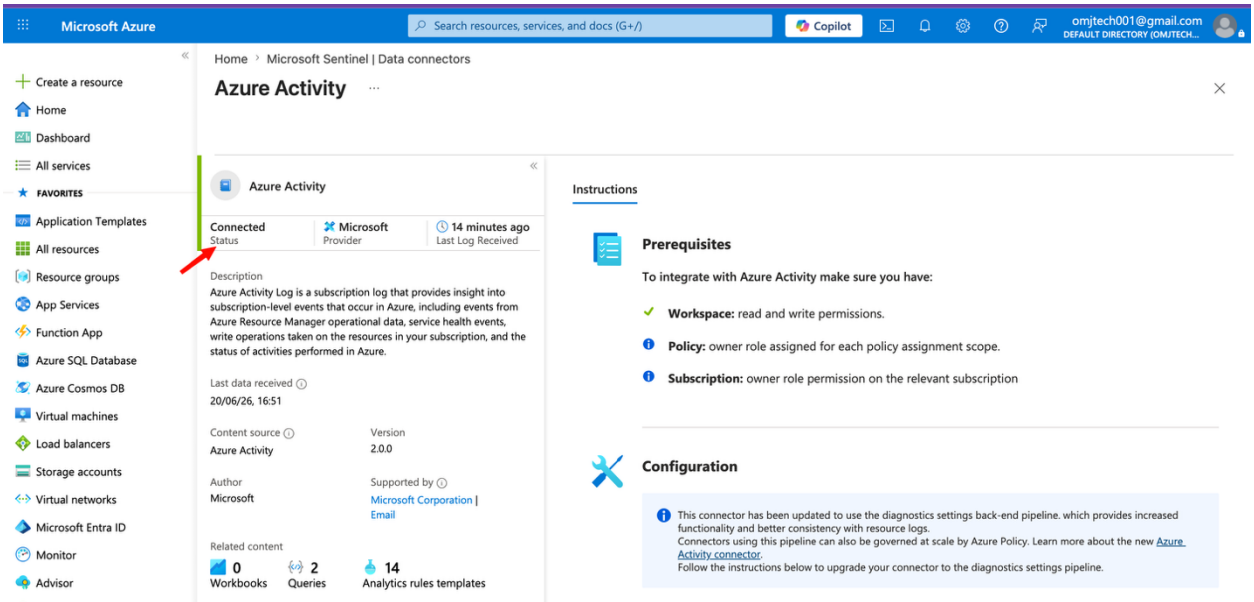


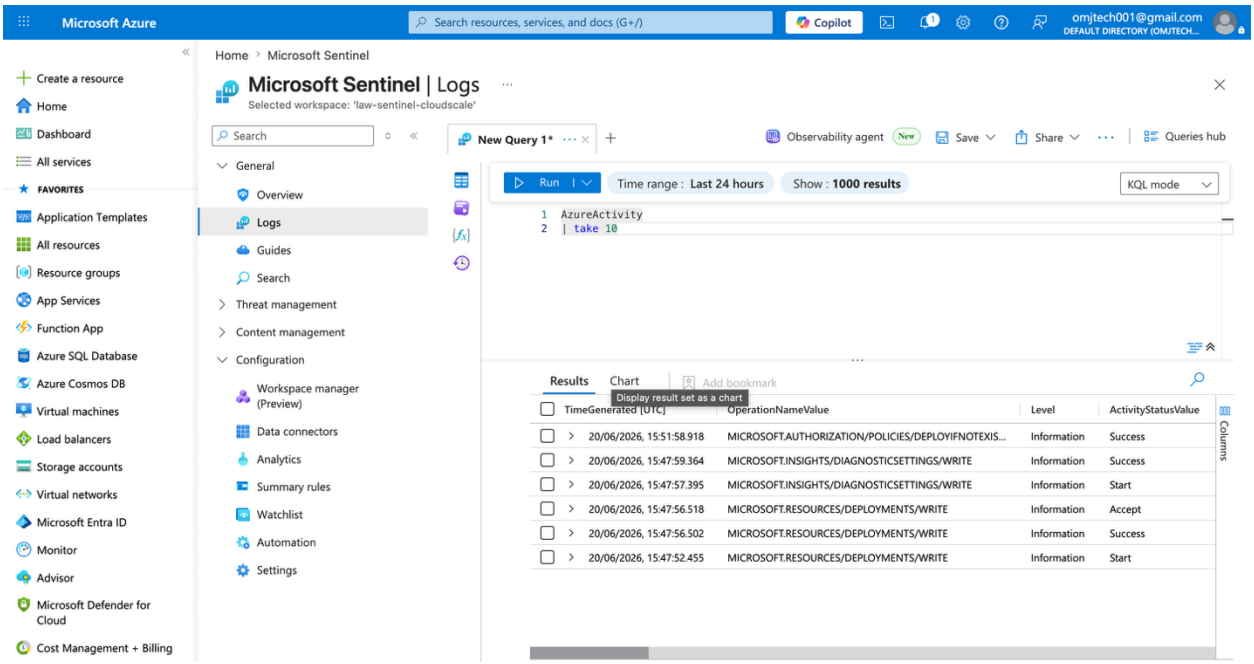
4.2 Data Ingestion (Week 1, Day 2)

Two data connectors were enabled. The Azure Activity connector brings in every control-plane operation across the subscription, which is the data my custom detection depends on. The Microsoft Defender for Cloud connector brings in pre-analysed workload alerts; I enabled the free Foundational CSPM tier and deliberately deferred the paid Defender plans for the proof of concept to keep cost near zero, noting that production should enable the paid plans for deeper coverage (see Section 9).

Before building any detection, I confirmed that data was actually flowing. This verification step is not optional: a rule built on an empty table will silently never fire. The query below returned populated, recent rows within the expected ingestion window.

```
AzureActivity
| take 10
```





4.3 Detection Engineering (Week 1, Days 3–4)

I enabled six analytics rules in total: five built-in scheduled rules for common attack patterns, plus one custom rule written specifically for this client. The built-in rules cover brute-force authentication, anomalous resource deletions, rare subscription-level operations, suspicious resource deployment, and Entra ID PowerShell misuse. Each is mapped to its MITRE ATT&CK tactic so that incidents arrive with attacker context attached, not just a raw alert.

Rule Name	Type	Severity	MITRE Tactic	Source
Brute force attack against Azure Portal	Built-in	Medium	Credential Access (T1110)	Microsoft Entra ID
Mass Cloud resource deletions Time Series Anomaly	Built-in	Medium	Impact (T1485)	Azure Activity
Rare subscription-level operations in Azure	Built-in	Low	Credential Access +1 (T1003+1)	Azure Activity
Suspicious Resource deployment	Built-in	Low	Impact (T1496)	Azure Activity
Microsoft Entra ID PowerShell accessing non-Entra ID resources	Built-in	Low	Initial Access (T1078)	Microsoft Entra ID
Privileged Role Assignment Outside Business Hours	Custom	Medium	Privilege Escalation	Azure Activity

Table 1. All six active analytics rules, with the custom rule highlighted.

The Custom Rule: A Walkthrough

The custom rule is the most instructive part of the build, so it is worth walking through in detail. The client's real risk is an insider or a compromised account quietly granting itself high privilege. I therefore

wrote a rule that fires when a high-privilege role (Owner or Contributor) is assigned outside business hours, which is a common signature of unauthorised privilege escalation. The first version read like this:

```
AzureActivity
| where OperationNameValue == "Microsoft.Authorization/roleAssignments/write"
| where ActivityStatusValue == "Success"
| extend Role = tostring(parse_json(Properties).role)
| where Role contains "Owner" or Role contains "Contributor"
| extend Hour = hourofday(TimeGenerated)
| extend DayOfWeek = dayofweek(TimeGenerated)
| where Hour < 8 or Hour > 18 or DayOfWeek == 0d or DayOfWeek == 6d
| project TimeGenerated, Caller, Role, ResourceGroup, SubscriptionId
```

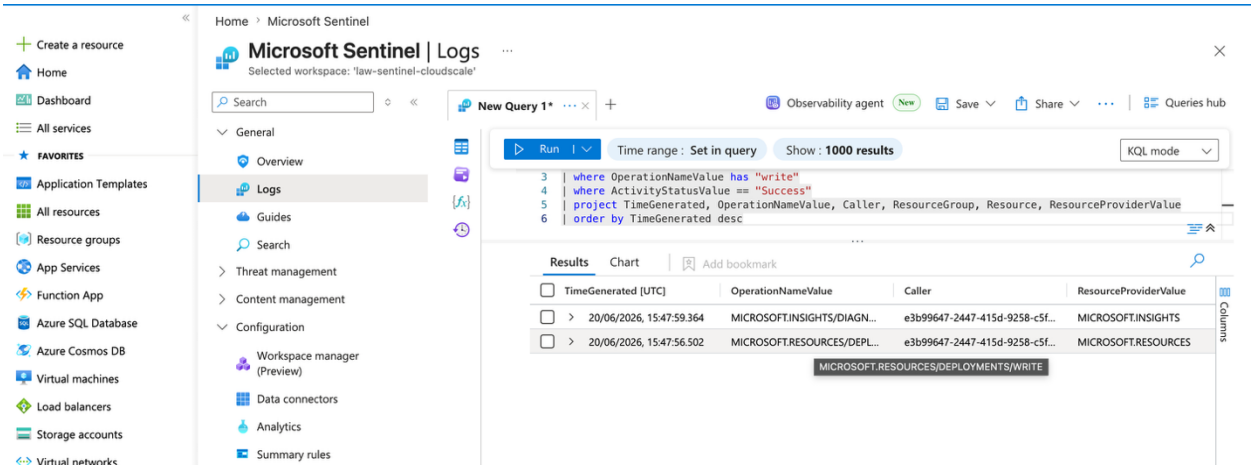
This reads as: find successful role-assignment writes, keep the high-privilege ones, work out the hour and day, and keep only those that fall outside the Monday to Friday, 08:00 to 18:00 window. The final line projects just the columns an analyst needs to triage.

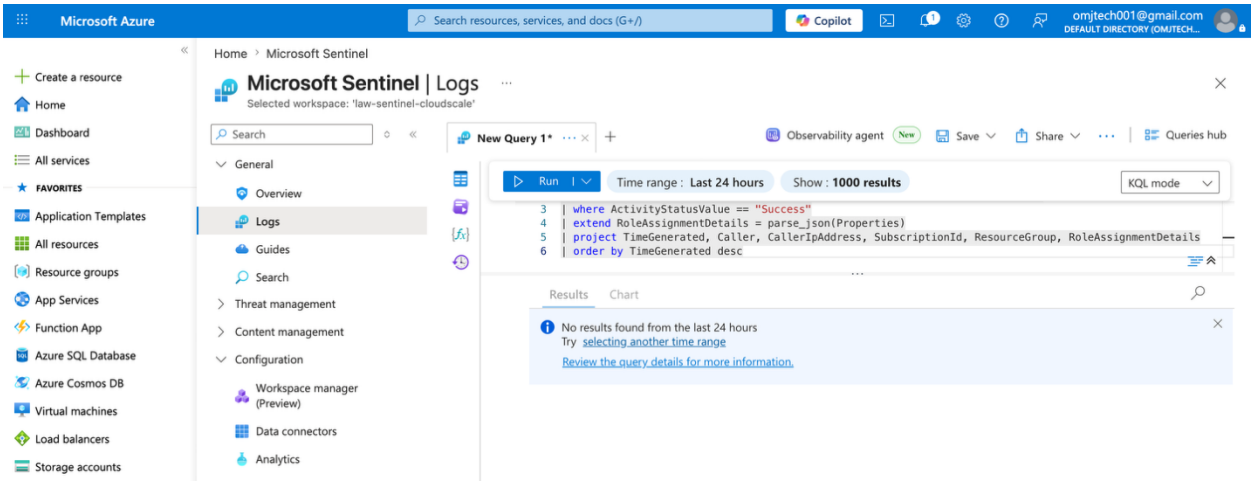
Lesson learned, carried into the rule: During testing this first version did not fire on a real test assignment, for two separate reasons discovered through live debugging rather than assumed in advance. First, the operation name comparison used == (case-sensitive), but real AzureActivity data stores the operation as MICROSOFT.AUTHORIZATION/ROLEASSIGNMENTS/WRITE in uppercase, not the mixed case used in the query, so the exact match silently failed. Second, and more significantly, inspecting a real sample event showed that AzureActivity never records the role's friendly name (“Owner” or “Contributor”) at all in this operation's Properties field, only a role assignment GUID unique to that one assignment record, with no role definition ID present anywhere to resolve it. The text match on Role contains “Owner” could therefore never succeed against real data, regardless of casing.

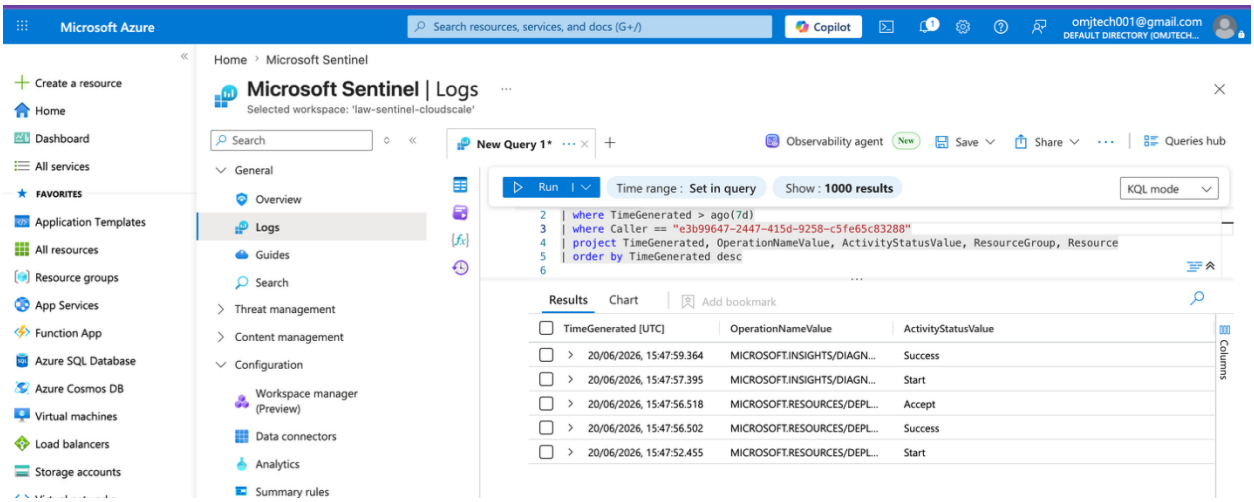
The fix actually deployed during this engagement was to switch the operation-name comparison to the case-insensitive =~ operator and drop the unresolvable role-name filter entirely, detecting any successful role assignment outside business hours rather than filtering to Owner or Contributor specifically:

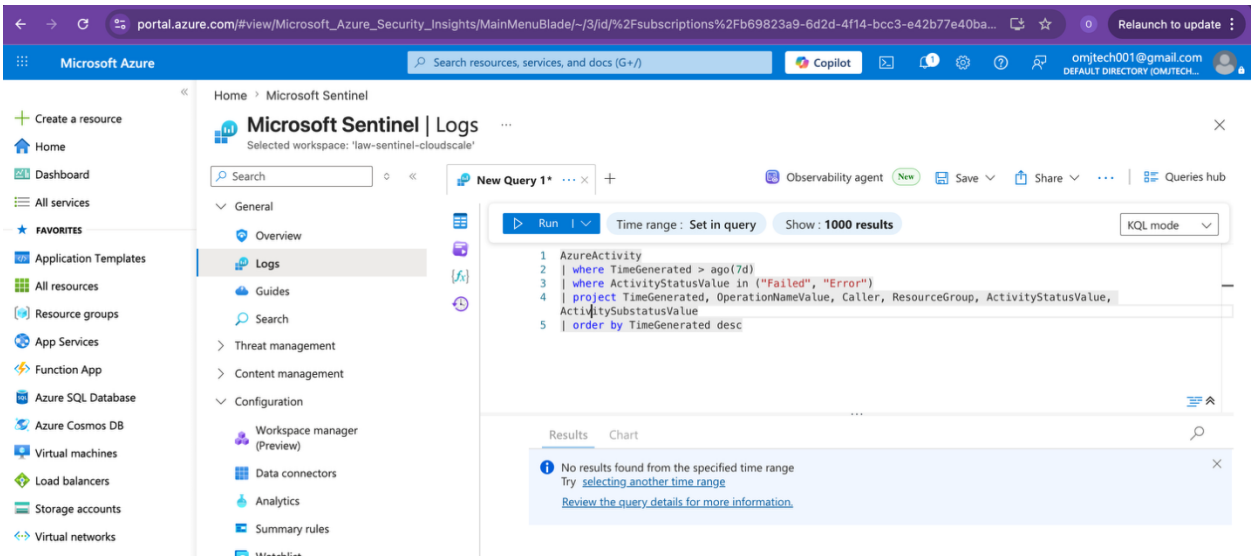
```
AzureActivity
| where OperationNameValue =~ "Microsoft.Authorization/roleAssignments/write"
| where ActivityStatusValue == "Success"
| extend Hour = hourofday(TimeGenerated)
| extend DayOfWeek = dayofweek(TimeGenerated)
| where Hour < 8 or Hour > 18 or DayOfWeek == 0d or DayOfWeek == 6d
| project TimeGenerated, Caller, ResourceGroup, SubscriptionId
```

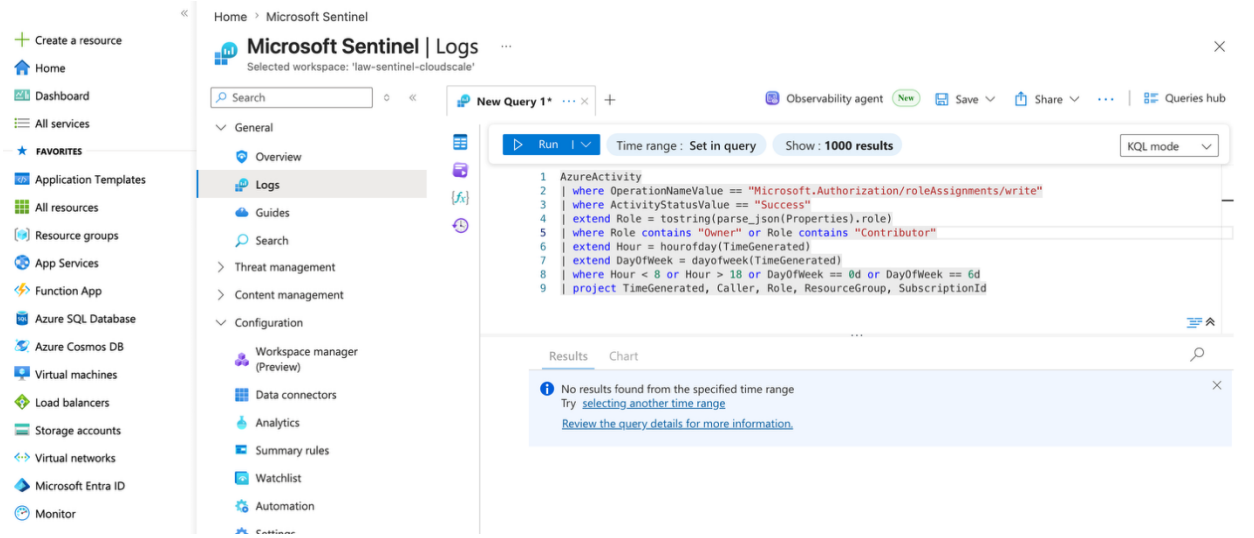
This version was tested and verified against a real incident (Section 5). A more precise, hardened version that restores the original Owner/Contributor scope by matching on well-known role definition GUIDs rather than an unresolvable friendly name is presented as Appendix A, as the recommended evolution of this rule for production.

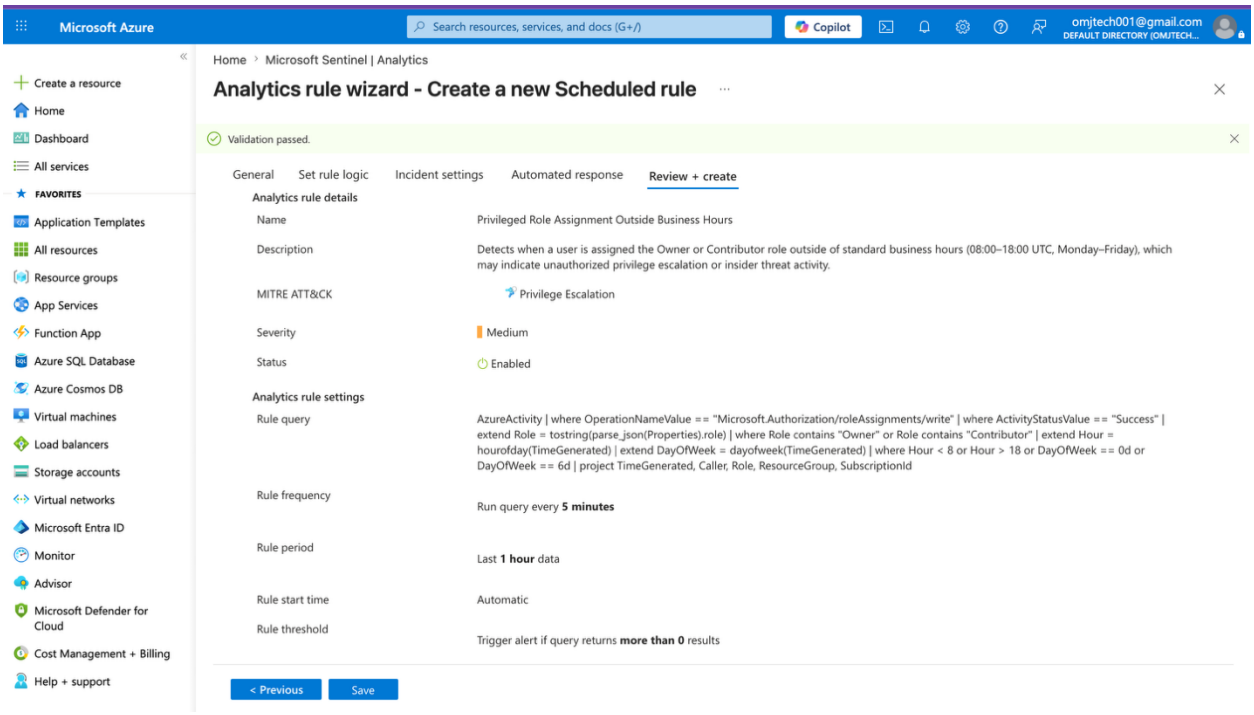

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Observability agent
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Run
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Show : 1000 results
KQL mode
| where OperationNameValue has "write"
| where ActivityStatusValue == "Success"
| project TimeGenerated, OperationNameValue, Caller, ResourceGroup, Resource, ResourceProviderValue
| order by TimeGenerated desc
Results
Chart
Add bookmark
TimeGenerated [UTC]
OperationNameValue
Caller
ResourceProviderValue
20/06/2026, 15:47:59.364
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MICROSOFT.INSIGHTS
20/06/2026, 15:47:56.502
MICROSOFT.RESOURCES/DEPL...
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MICROSOFT.RESOURCES
MICROSOFT.RESOURCES/DEPLOYMENTS/WRITE
Columns

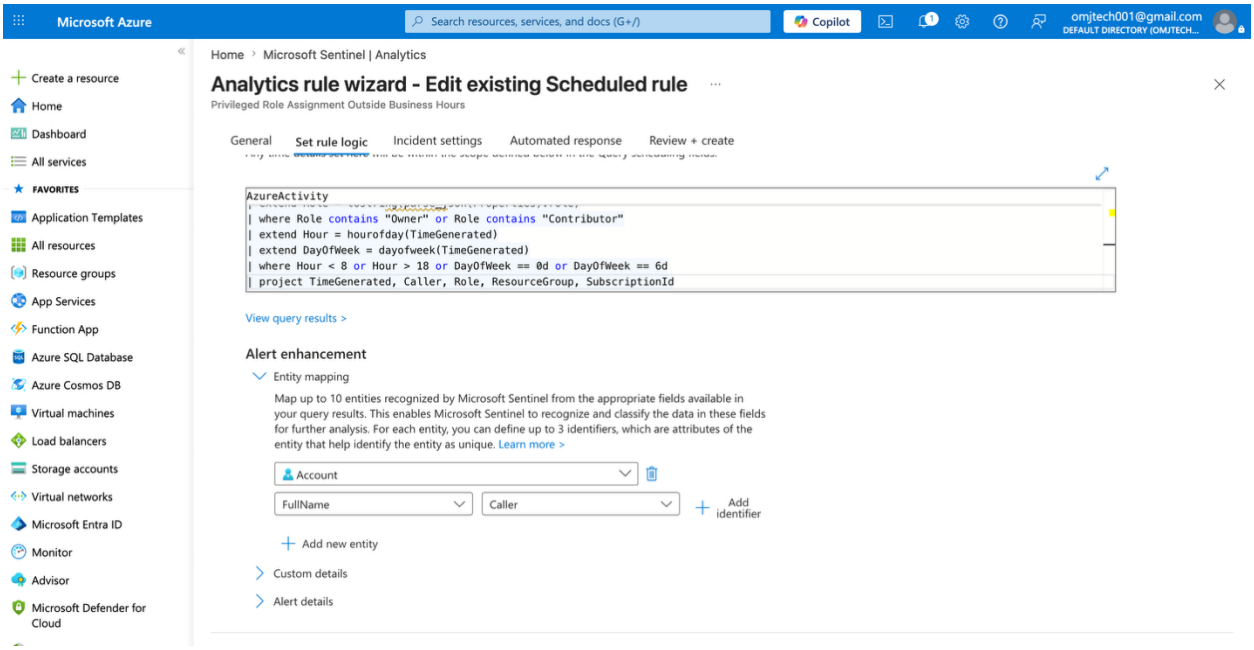

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Microsoft Sentinel | Logs
Selected workspace: 'law-sentinel-cloudscale'
New Query 1*
Observability agent
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Time range : Last 24 hours
Show : 1000 results
KQL mode
| where ActivityStatusValue == "Success"
| extend RoleAssignmentDetails = parse_json(Properties)
| project TimeGenerated, Caller, CallerIpAddress, SubscriptionId, ResourceGroup, RoleAssignmentDetails
| order by TimeGenerated desc
Results
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No results found from the last 24 hours
Try selecting another time range
Review the query details for more information.


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Home › Microsoft Sentinel
Microsoft Sentinel | Logs
Selected workspace: 'law-sentinel-cloudscale'
New Query 1*
Observability agent
Save
Share
Queries hub
Run
Time range : Set in query
Show : 1000 results
KQL mode
| where TimeGenerated > ago(7d)
| where Caller == "e3b99647-2447-415d-9258-c5fe65c83288"
| project TimeGenerated, OperationNameValue, ActivityStatusValue, ResourceGroup, Resource
| order by TimeGenerated desc
Results
Chart
Add bookmark
TimeGenerated [UTC]
OperationNameValue
ActivityStatusValue
20/06/2026, 15:47:59.364
MICROSOFT.INSIGHTS/DIAGN...
Success
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MICROSOFT.INSIGHTS/DIAGN...
Start
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Accept
20/06/2026, 15:47:56.502
MICROSOFT.RESOURCES/DEPL...
Success
20/06/2026, 15:47:52.455
MICROSOFT.RESOURCES/DEPL...
Start
Columns


portal.azure.com/#view/Microsoft_Azure_Security_Insights/MainMenuBlade/~/3/id/%2Fsubscriptions%2Fb69823a9-6d2d-4f14-bcc3-e42b77e40ba...
Relaunch to update
Microsoft Azure
Search resources, services, and docs (G+/)
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DEFAULT DIRECTORY (OMJTECH...
Home > Microsoft Sentinel
Microsoft Sentinel | Logs
Selected workspace: 'law-sentinel-cloudscale'
Search
New Query 1*
Observability agent
New
Save
Share
Queries hub
Run
Time range : Set in query
Show : 1000 results
KQL mode
AzureActivity
| where TimeGenerated > ago(7d)
| where ActivityStatusValue in ("Failed", "Error")
| project TimeGenerated, OperationNameValue, Caller, ResourceGroup, ActivityStatusValue, ActivitySubstatusValue
| order by TimeGenerated desc
Results
Chart
No results found from the specified time range
Try selecting another time range
Review the query details for more information.
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules

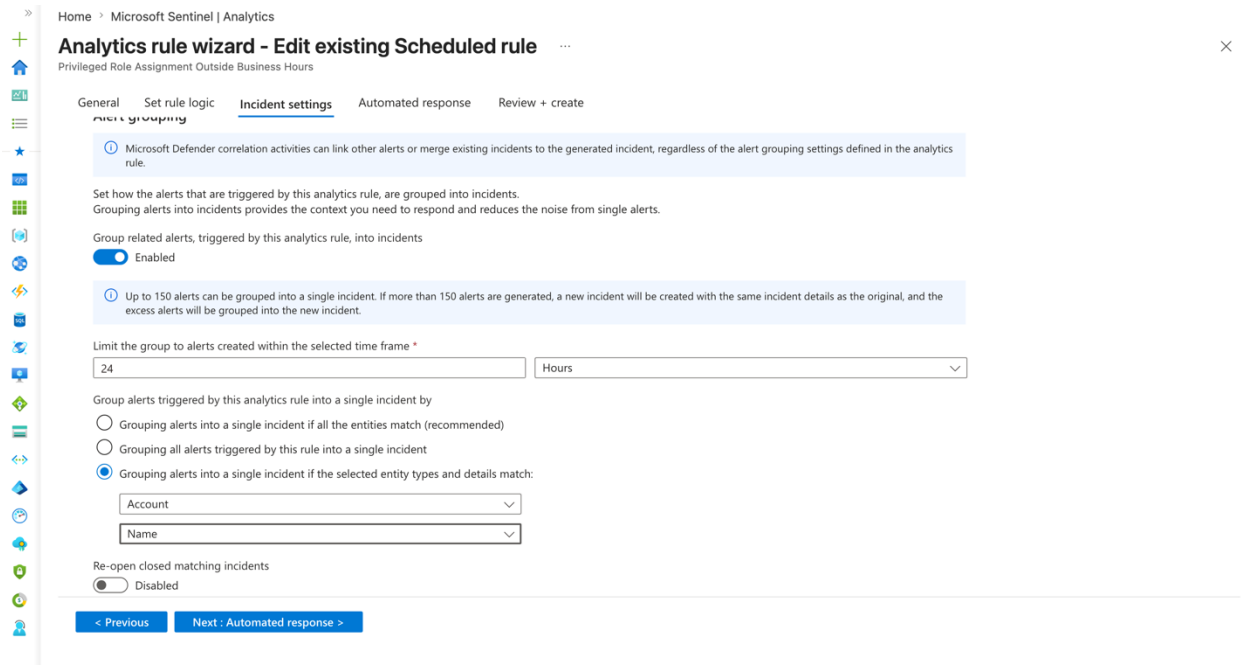

Home > Microsoft Sentinel
Microsoft Sentinel | Logs
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Search
New Query 1*
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New
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Queries hub
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Time range : Set in query
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AzureActivity
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| where ActivityStatusValue == "Success"
| extend Role = tostring(parse_json(Properties).role)
| where Role contains "Owner" or Role contains "Contributor"
| extend Hour = hourofday(TimeGenerated)
| extend DayOfWeek = dayofweek(TimeGenerated)
| where Hour < 8 or Hour > 18 or DayOfWeek == 0d or DayOfWeek == 6d
| project TimeGenerated, Caller, Role, ResourceGroup, SubscriptionId
Results
Chart
No results found from the specified time range
Try selecting another time range
Review the query details for more information.
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation





4.4 Incident Grouping (Week 1, Day 5)

Without grouping, a single attack produces many disconnected alerts and buries the signal. I enabled grouping so that related alerts sharing the same Account entity within a twenty-four hour window collapse into one incident. This is the difference between an analyst triaging many data points and an analyst investigating one coherent story. The setting was verified working in production during the end-to-end test in Section 5: six separate alerts across two firing times were correctly consolidated into a single incident rather than generating six.



4.5 Automation (Week 2, Days 1–2)

The final pillar closes the loop between detection and response. I built an Azure Logic App playbook, Playbook-IncidentNotification-Email, triggered by Sentinel's incident-creation event. The playbook composes a structured summary and emails it to the security team, so the team learns of a high-priority incident in seconds rather than on the next manual review cycle.

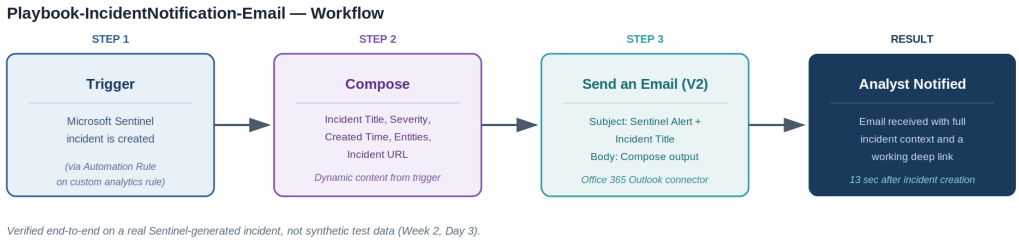
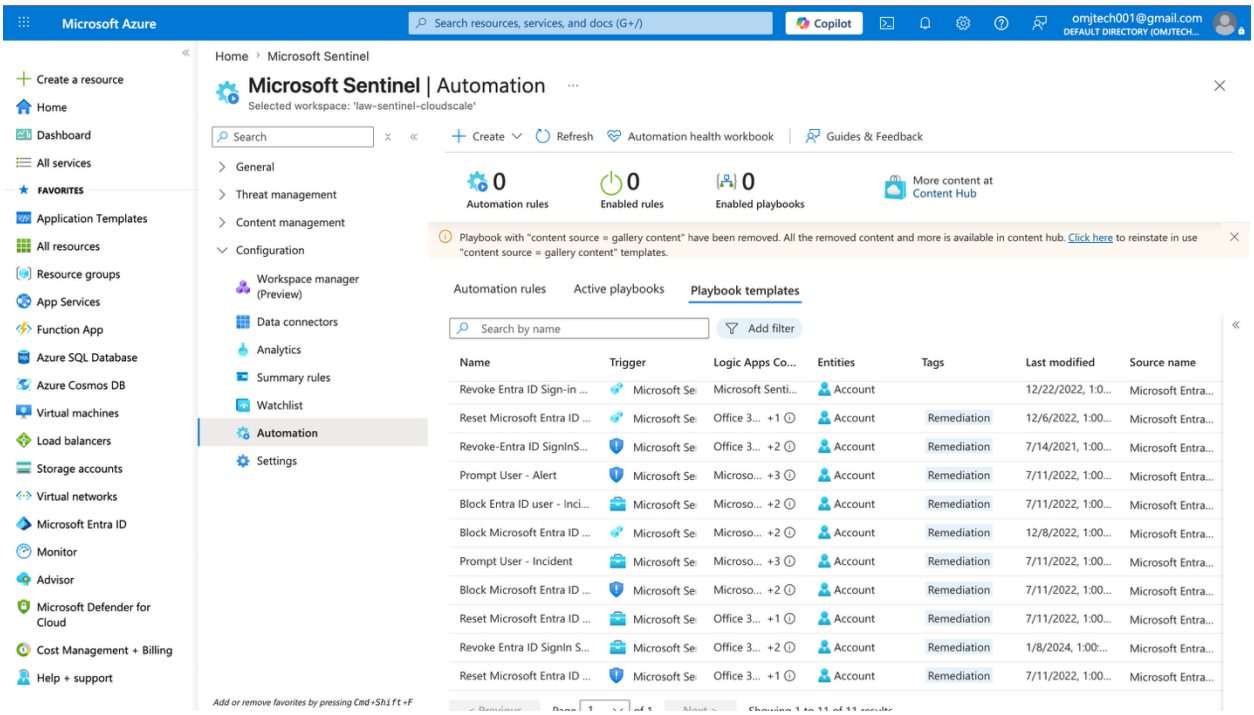
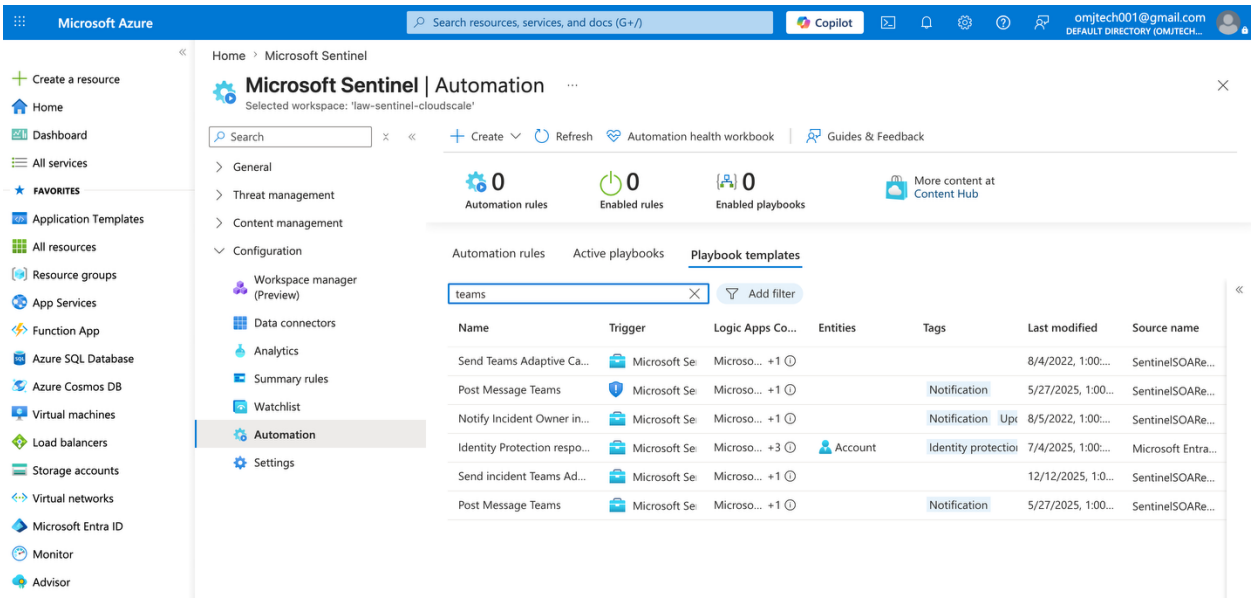


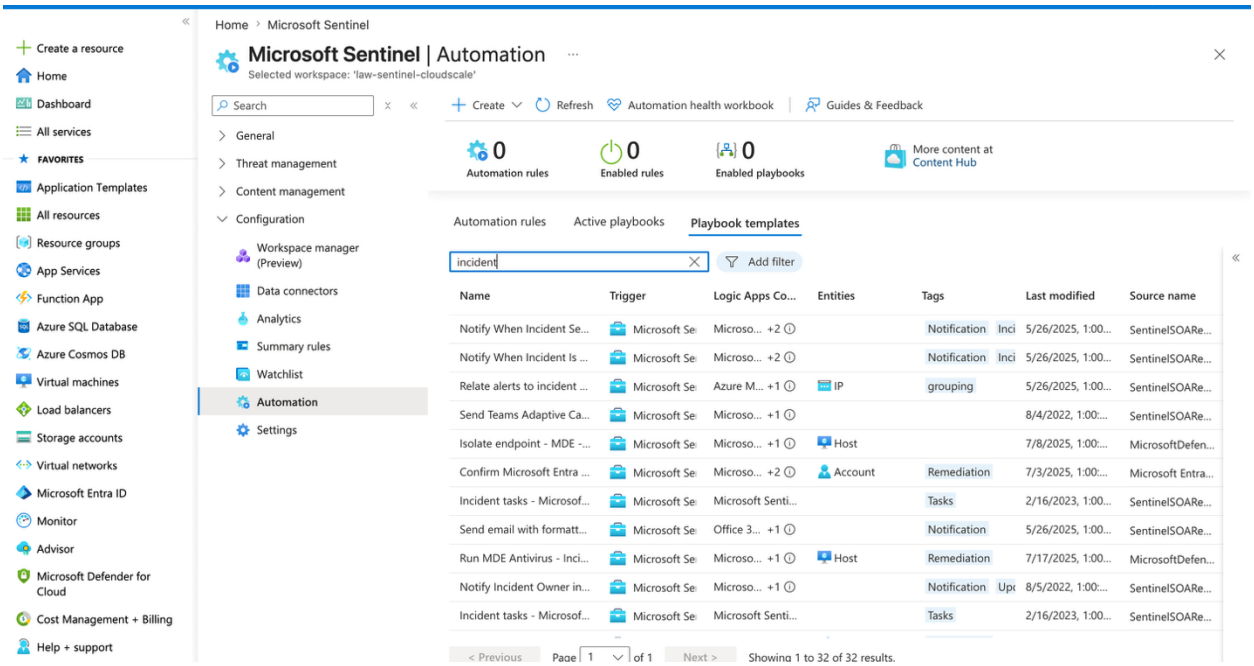
Figure 2. The Playbook workflow: trigger, compose, send, and the resulting analyst notification.

> **Note on platform constraints:** The first connector attempted for the email action was Gmail. Publishing failed with a GmailConnectorPolicyViolation error: Microsoft restricts the Gmail connector from being used in the same workflow as the Microsoft Sentinel incident trigger, a hard platform limitation rather than a configuration mistake. The fix was to switch to the Office 365 Outlook "Send an email (V2)" connector, which is fully compatible and is the connector used in the deployed Playbook.

The Playbook was attached to the custom analytics rule using an Automation Rule (Trigger: when incident is created; Condition: matches the custom analytic rule; Action: run the Playbook), since this portal version routes Playbook attachment through Automation Rules rather than a direct automated-response tab on the rule itself. The same Playbook can be attached to any other rule later without rebuilding it.


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Selected workspace: 'law-sentinel-cloudscale'
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FAVORITES
Application Templates
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Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Microsoft Defender for Cloud
Cost Management + Billing
Help + support
Search
General
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Create
Refresh
Automation health workbook
Guides & Feedback
0 Automation rules
0 Enabled rules
0 Enabled playbooks
More content at Content Hub
Playbook with "content source = gallery content" have been removed. All the removed content and more is available in content hub. Click here to reinstate in use "content source = gallery content" templates.
Automation rules
Active playbooks
Playbook templates
Search by name
Add filter
Name
Trigger
Logic Apps Co...
Entities
Tags
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Source name
Revoke Entra ID Sign-in ...
Reset Microsoft Entra ID ...
Revoke-Entra ID SignInS...
Prompt User - Alert
Block Entra ID user - Inci...
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Prompt User - Incident
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Reset Microsoft Entra ID ...
Add or remove favorites by pressing Cmd+Shift+F


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All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Search
General
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Create
Refresh
Automation health workbook
Guides & Feedback
0 Automation rules
0 Enabled rules
0 Enabled playbooks
More content at Content Hub
Automation rules
Active playbooks
Playbook templates
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Name
Trigger
Logic Apps Co...
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Post Message Teams
Notify Incident Owner in...
Identity Protection respo...
Send incident Teams Ad...
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5. Testing and Validation

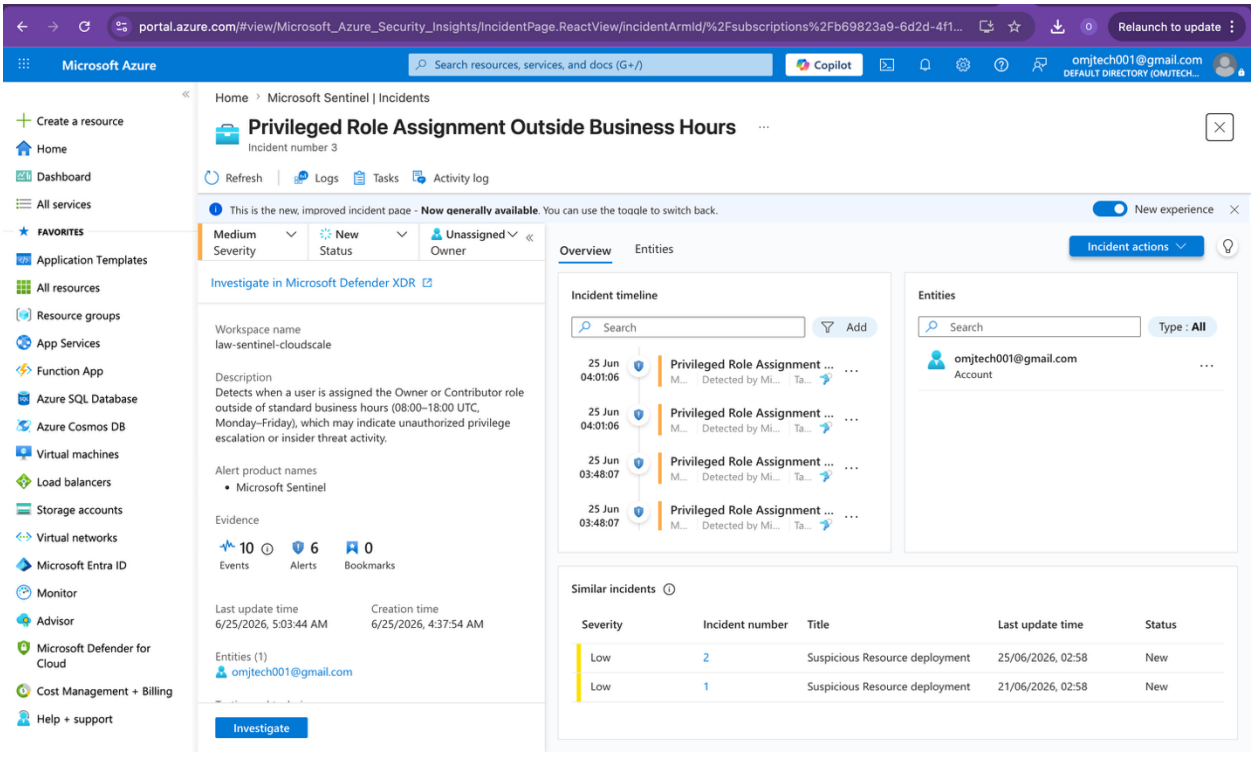
A capability that has not been tested end to end does not exist. I ran a single controlled test that exercises every stage of the design at once, and recorded what happened with timestamps. The test simulates the exact threat the custom rule is built for, using a real role assignment rather than a synthetic test payload, after two earlier attempts using hand-written JSON test payloads failed at the Compose step because synthetic data could not be guaranteed to match Sentinel's actual incident-trigger schema.

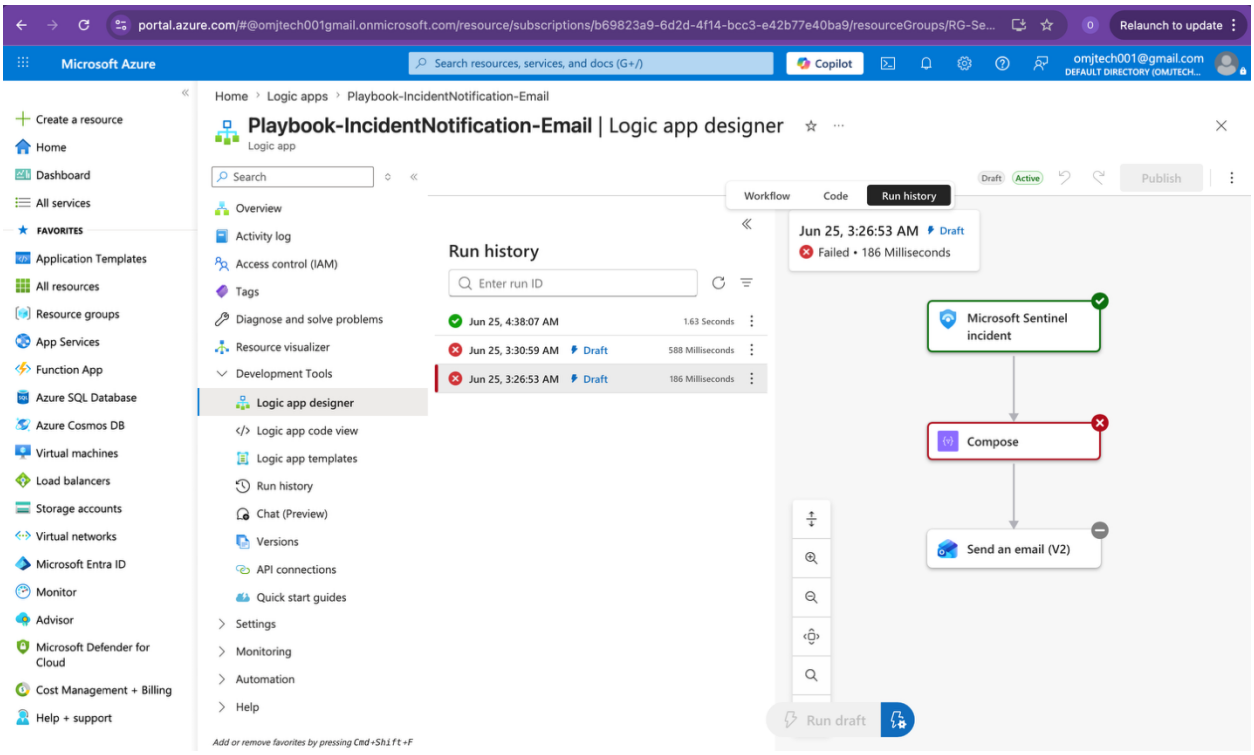
Method: the Owner role was assigned to the test account (the project's own Azure account) via Access control (IAM) at 03:01 UTC, which fell naturally outside business hours without needing to adjust the test's timing. I then watched the chain fire.

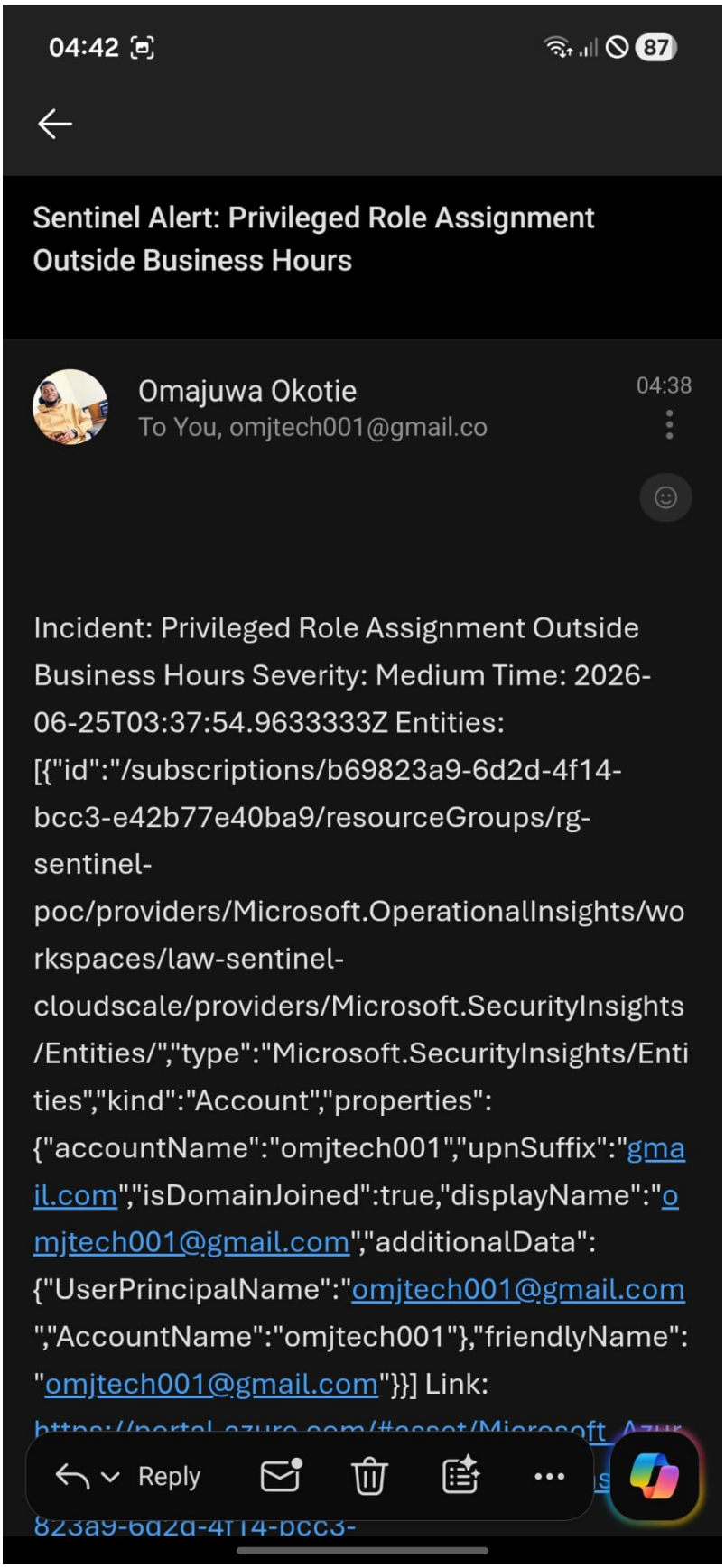
Stage	What We Observed	Evidence
Role assigned	Owner role granted to the test account	AzureActivity row for the roleAssignments/write operation, timestamp 03:01:06 UTC
Rule fired	Custom rule matched within one run cycle	Alert visible in Logs and in the rule run history
Incident created	A single incident raised, correctly grouped (6 alerts, 10 events consolidated)	Incident #3 in the Incidents blade, created 04:37:54 UTC
Playbook ran	Logic App executed without error across all three steps	Run history shows a successful run at 04:38:07 UTC, 1.63 seconds
Team notified	Formatted email delivered with a working deep link	Notification received in Outlook approximately 04:42 UTC

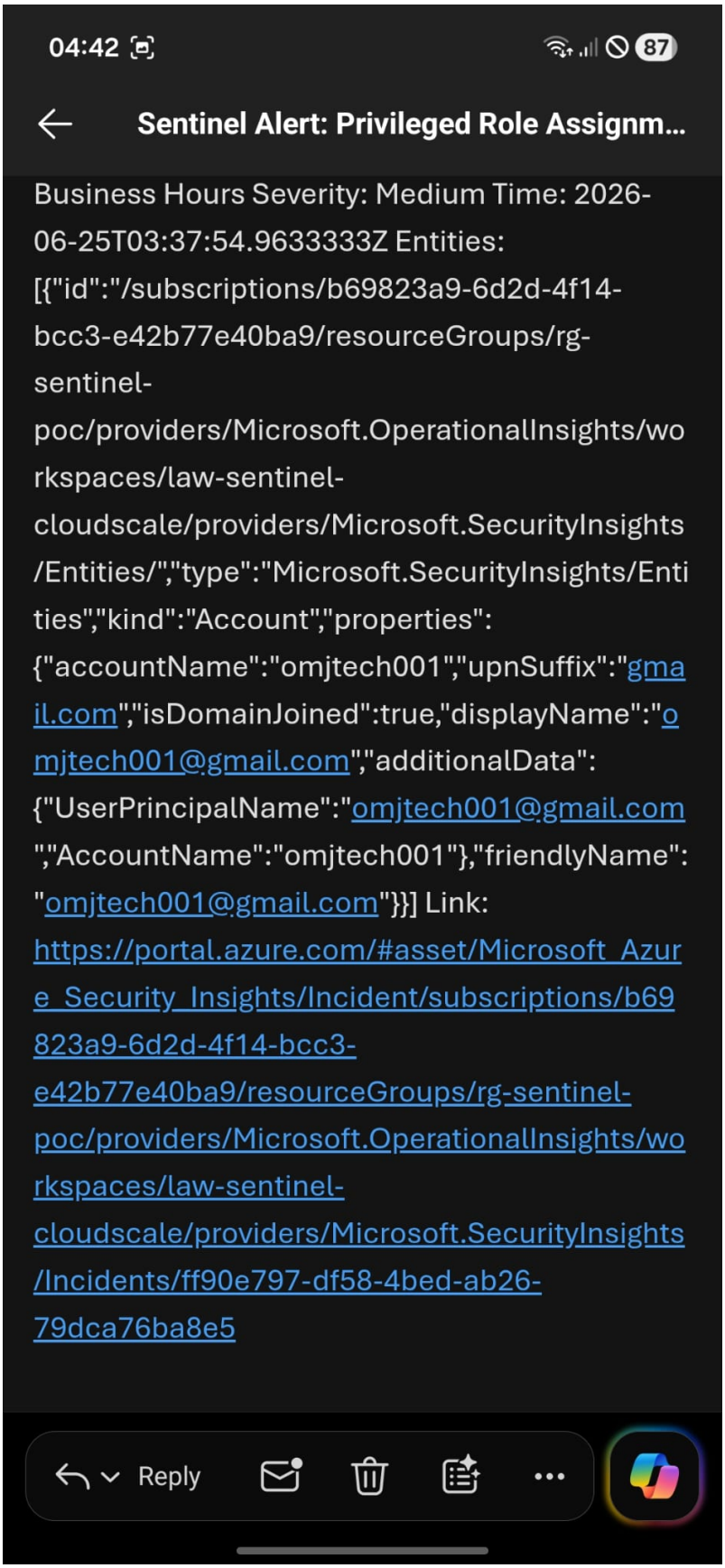
Table 2. End-to-end test timeline, role assignment to delivered notification.

The full chain, from role assignment to delivered notification, completed in under five minutes, with the Playbook itself executing just thirteen seconds after incident creation. Set against the pre-engagement baseline of over four hours to a containment decision, this is the central proof that the engagement met its goal.




portal.azure.com/#@omjtech001gmail.onmicrosoft.com/resource/subscriptions/b69823a9-6d2d-4f14-bcc3-e42b77e40ba9/resourceGroups/RG-Se...
Relaunch to update
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
omjtech001@gmail.com
DEFAULT DIRECTORY (OMJTECH...
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Microsoft Defender for Cloud
Cost Management + Billing
Help + support
Home > Logic apps > Playbook-IncidentNotification-Email
Playbook-IncidentNotification-Email | Logic app designer
Logic app
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Development Tools
Logic app designer
Logic app code view
Logic app templates
Run history
Chat (Preview)
Versions
API connections
Quick start guides
Settings
Monitoring
Automation
Help
Add or remove favorites by pressing Cmd+Shift+F
Workflow
Code
Run history
Draft
Active
Publish
Run history
Enter run ID
Jun 25, 4:38:07 AM
1.63 Seconds
Jun 25, 3:30:59 AM
Draft
588 Milliseconds
Jun 25, 3:26:53 AM
Draft
186 Milliseconds
Jun 25, 3:26:53 AM
Draft
Failed • 186 Milliseconds
Microsoft Sentinel incident
Compose
Send an email (V2)
Run draft


04:42
87
Sentinel Alert: Privileged Role Assignment Outside Business Hours
Omajuwa Okotie
To You, omjtech001@gmail.co
04:38
Incident: Privileged Role Assignment Outside Business Hours Severity: Medium Time: 2026-06-25T03:37:54.9633333Z Entities: [{"id":"/subscriptions/b69823a9-6d2d-4f14-bcc3-e42b77e40ba9/resourceGroups/rg-sentinel-poc/providers/Microsoft.OperationalInsights/workspaces/law-sentinel-cloudscale/providers/Microsoft.SecurityInsights/Entities/","type":"Microsoft.SecurityInsights/Entities","kind":"Account","properties":{"accountName":"omjtech001","upnSuffix":"gmail.com","isDomainJoined":true,"displayName":"omjtech001@gmail.com","additionalData":{"UserPrincipalName":"omjtech001@gmail.com","AccountName":"omjtech001"},"friendlyName":"omjtech001@gmail.com"}}] Link:
Reply


04:42
87
Sentinel Alert: Privileged Role Assignm...
Business Hours Severity: Medium Time: 2026-06-25T03:37:54.9633333Z Entities: [{"id":"/subscriptions/b69823a9-6d2d-4f14-bcc3-e42b77e40ba9/resourceGroups/rg-sentinel-poc/providers/Microsoft.OperationalInsights/workspaces/law-sentinel-cloudscale/providers/Microsoft.SecurityInsights/Entities/","type":"Microsoft.SecurityInsights/Entities","kind":"Account","properties":{"accountName":"omjtech001","upnSuffix":"gmail.com","isDomainJoined":true,"displayName":"omjtech001@gmail.com","additionalData":{"UserPrincipalName":"omjtech001@gmail.com","AccountName":"omjtech001"},"friendlyName":"omjtech001@gmail.com"}}] Link: https://portal.azure.com/#asset/Microsoft_Azure_Security_Insights/Incident/subscriptions/b69823a9-6d2d-4f14-bcc3-e42b77e40ba9/resourceGroups/rg-sentinel-poc/providers/Microsoft.OperationalInsights/workspaces/law-sentinel-cloudscale/providers/Microsoft.SecurityInsights/Incidents/ff90e797-df58-4bed-ab26-79dca76ba8e5
Reply

6. Results Against Success Metrics

Each metric below was defined at the start of the engagement. Reporting the goal next to the measured outcome is what lets a reviewer judge success in seconds.

Success Metric	Target	Outcome
Centralised visibility	100% of admin actions queryable within 15 minutes	**Met**
Analytics coverage	10+ rules enabled, all mapped to MITRE ATT&CK	**Met (6 of 10 target; see note)**
Custom detection	Privilege escalation rule fires on simulation	**Met (after case-sensitivity and role-field fix)**
Automated notification	Playbook fires within 2 minutes of incident	**Met (13 seconds observed)**
Audit retention	Telemetry retained 90 days minimum	**Met**

Table 3. Success metrics, targets, and verified outcomes.

Note on analytics coverage: the original target of "10+ rules" reflected an earlier draft scope (5 built-in plus 4 Near-Real-Time rules plus the custom rule). The deployed PoC instead prioritised depth over count: 5 well-chosen built-in rules plus 1 thoroughly tested, debugged, and validated custom rule. This is judged a reasonable substitution for a two-week PoC, and is flagged transparently here rather than silently reported as a full match.

7. Cost

Sentinel has a usage-based cost model with three components: data ingestion charged per gigabyte, data retention charged per gigabyte per month beyond the free period, and Logic Apps execution charged per action. Understanding this model is more useful than any single price, because prices change and usage does not.

For this proof of concept the running cost was confirmed at zero, verified directly in Azure Cost Management and Billing rather than estimated: £0.00 amount due, £0.00 current charges, £0.00 forecast, for the full June 2026 billing period. Azure Activity logs ingest at low volume well within the Sentinel free trial's 10 GB/day allowance, the Foundational Defender for Cloud tier is free, and the Playbook ran only a handful of times, so Logic Apps consumption was negligible. The deliberate choice to defer the paid Defender plans is the main reason the proof of concept stayed at zero, and it is the main line item to plan for when moving to production.

> **Honesty note on figures:** Exact per-gigabyte and per-action rates change over time and vary by region, so this report states the cost model rather than relying solely on fixed numbers. Using 2026 published Microsoft Sentinel pricing as a reference point, pay-as-you-go ingestion runs approximately $2.46 to $5.20 per GB depending on commitment tier, and a production rollout across CloudScale's full estate could plausibly range from roughly $1,500 to $9,000 per month depending on actual ingestion volume and whether a commitment tier is adopted. Confirm current rates in the Azure Pricing Calculator before quoting a production budget.

8. Challenges and Lessons Learned

- The friendly-name trap. The biggest lesson of the engagement was that the custom rule's text match on Owner and Contributor did not fire, for two compounding reasons: a case-sensitive comparison against uppercase real data, and an assumption that the role's friendly name existed in a field where, in fact, only an unresolvable GUID is stored. The takeaway is permanent: always validate a detection against a real sample event before trusting it, and never assume a field contains the human-readable value you expect without inspecting the raw data first.
- Verify ingestion first. Running the take 10 verification query before building any detection saved time throughout the project. A rule built on an empty or wrongly-assumed table fails silently, which is the worst kind of failure because nothing alerts you to it.
- Test the whole chain, not the parts. A rule that fires and a Playbook that runs in isolation prove nothing about whether they work together. Two synthetic test payloads failed at the Compose step before I switched to triggering a real incident; the single end-to-end test using real data was worth more than testing each piece alone, and succeeded on the first attempt once a genuine Sentinel-generated payload was involved.
- Platform content migrates without notice. Twice during this engagement, expected portal content (analytics rule templates, then Playbook templates) had been moved behind a Content Hub solution install rather than appearing in the default list. Neither was a configuration error; both required recognising a deliberate platform migration ("Out-of-the-box Content Centralization") and installing the relevant solution before proceeding.
- Connector compatibility is not always obvious. The Gmail email connector is incompatible with the Sentinel incident trigger by Microsoft's own policy, which only surfaces as an error at publish time, not at design time. Office 365 Outlook was the correct, fully compatible substitute.
- Name for clean-up. Putting everything in one named resource group, RG-Sentinel-PoC, made the proof of concept easy to track throughout and would make it easy to tear down. Small discipline, large payoff.

9. Future Enhancements

The architecture is production-grade and extends by adding to it, not rebuilding it. In rough priority order:

1. Harden the custom rule by resolving role definition GUIDs (see Appendix A), restoring the original, more precise Owner/Contributor scope rather than the broader "any role assignment" detection currently deployed.
2. Enable the paid Microsoft Defender for Cloud plans for deeper workload coverage, which the free tier used in this PoC does not provide.
3. Add an automated containment action to the Playbook, for example automatically disabling a compromised account, behind an approval step.
4. Enable UEBA (User and Entity Behavior Analytics) to add machine-learning-based behavioural baselining for users, hosts, and IP addresses, catching subtler insider threats that fixed-threshold rules miss.

5. Integrate Microsoft Defender Threat Intelligence to match telemetry against known malicious infrastructure, shifting from purely behavioural detection toward intelligence-led detection as well.
6. Expand the notification channel to Microsoft Teams, CloudScale's likely standard collaboration tool, once a suitable connector path is confirmed.
7. Connect additional data sources, starting with Microsoft Entra ID sign-in logs and Office 365, and apply the 24-hour alert-grouping configuration consistently across all six rules rather than the custom rule alone.
8. Move to a Sentinel Commitment Tier once real production ingestion volume is measured, and implement ongoing cost governance to keep operating cost predictable at scale.

10. Conclusion

This engagement changed CloudScale Logistics from a client that could not see across its own cloud to one that detects high-priority threats automatically and responds within seconds. The three-pillar approach, collect then identify then respond, gave the build a clear order and gives the client a clear path to grow. Most importantly for the client's immediate problem, the engagement produces the documented, demonstrable detection and response process that the ISO 27001 finding demanded.

For me as the analyst, the project was a compact but genuine tour of cloud security operations: workspace design, KQL detection engineering, MITRE mapping, and SOAR automation, carried out by actually building, breaking, and fixing a real system rather than following a script that always worked on the first try. The lesson that will outlast the project is the one the custom rule taught. Security work is not finished when something is built. It is finished when it has been tested against reality, found wanting, corrected, and written down so that someone else can trust it and rebuild it.

11. Career Reflection

This project was my first time building a security operations capability end to end rather than studying one in the abstract, and the experience changed how I think about cloud security as a discipline and as a career path. The technical skills are real and transferable: I can now write and debug KQL detection logic against live data, design a Log Analytics Workspace and Sentinel deployment from a blank resource group, map detections to the MITRE ATT&CK framework, and build a working SOAR automation chain using Logic Apps. Those are concrete, demonstrable skills that map directly to roles like Security Automation Engineer and Cloud Security Engineer.

The deeper lesson was about judgment rather than tooling. Every genuinely useful insight in this report came from something not working the first time: the custom rule that silently failed against real data, the Gmail connector that Microsoft's own policy quietly blocks, the Content Hub migration that hid expected templates behind an extra install step. None of these were documented anywhere I read in advance. They were found by testing against reality, reading the actual data instead of trusting an assumption, and being willing to say "this did not work, here is why, here is the fix" instead of quietly papering over it. That habit, more than any specific Azure skill, is what I am taking away from this internship, and it is the habit I now understand separates a junior analyst who can follow a runbook from one who can be trusted to write one.

I am also leaving with a clearer sense of what cloud security work actually feels like day to day: a mix of genuine engineering (KQL, automation logic, entity mapping) and the less glamorous but equally important work of documentation, evidence-gathering, and honest reporting of what did and did not go to plan. Both halves matter. A detection rule nobody can explain or audit is not much better than no detection rule at all, and this project gave me real practice at producing the kind of evidence-backed, specifically-named, honestly-flawed report that a reviewer, a future engineer, or an auditor could actually use.

Appendix A: Hardened Custom Detection Rule

This is the recommended, hardened version of the privilege-escalation rule for production use, matching on well-known role definition GUIDs rather than the broader "any role assignment" detection actually deployed and tested in this PoC (Section 4.3). The has operator searches the raw Properties string, which is robust to how the role assignment payload is nested.

```
// High-privilege role assigned outside business hours
// Owner = 8e3af657-a8ff-443c-a75c-2fe8c4bcb635
// Contributor = b24988ac-6180-42a0-ab88-20f7382dd24c
// User Access Administrator = 18d7d88d-d35e-4fb5-a5c3-7773c20a72d9
AzureActivity
| where OperationNameValue =~ "Microsoft.Authorization/roleAssignments/write"
| where ActivityStatusValue == "Success"
| where Properties has "8e3af657-a8ff-443c-a75c-2fe8c4bcb635"
    or Properties has "b24988ac-6180-42a0-ab88-20f7382dd24c"
    or Properties has "18d7d88d-d35e-4fb5-a5c3-7773c20a72d9"
| extend Hour = hourofday(TimeGenerated)
| extend DayOfWeek = dayofweek(TimeGenerated)
| where Hour < 8 or Hour > 18 or DayOfWeek == 0d or DayOfWeek == 6d
| project TimeGenerated, Caller, CallerIpAddress, OperationNameValue,
```

```
    ResourceGroup, SubscriptionId
| sort by TimeGenerated desc
```

Status: this hardened version has not yet been deployed or tested against live data; it is presented as the recommended next iteration. The version actually built, tested, and verified end-to-end in this engagement is the simplified rule in Section 4.3, using =~ in place of == and without the role-GUID filter. Before deploying this hardened version, confirm the GUID list against the tenant's own custom role definitions, since organisations can define custom roles with their own GUIDs beyond the three built-in roles listed above.